

1) $\dot{x} = 5 - r e^{-x^2}$ Charles Michael Hayes

a.

$$r_c = 5$$

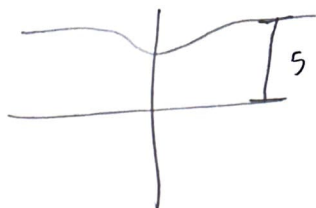
@ large $|x|$, $\dot{x} \approx 5$

@ $x=0$, $\dot{x} = 5 - r$

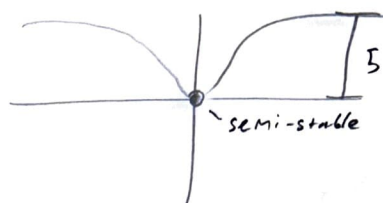
$\Rightarrow$ @ $r = r_c = 5$, $\dot{x}(x=0) = 0$

@ $r > r_c$, $\dot{x} > 0$

$$r < r_c$$



$$r = r_c$$

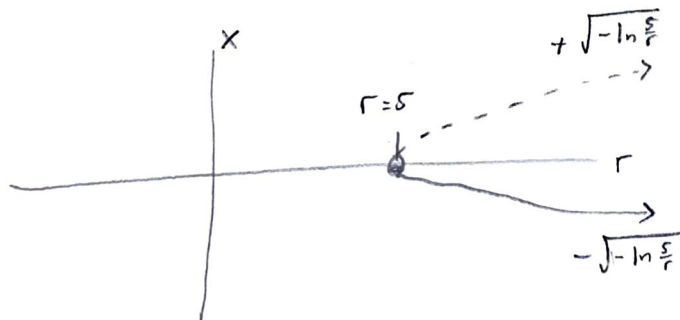


$\hookrightarrow 0 = 5 - r e^{-x^2}$ has 2 solutions



b. The bifurcation that occurs at $r = r_c = 5$ is a saddle-node (or blue sky) bifurcation. Since the Taylor expansion of e^{-x^2} is $1 - x^2$, the behavior of $\dot{x} = f(x) = 5 - r e^{-x^2}$ is that of approximately $\dot{x} \approx x^2$, akin to the quadratic normal form of the saddle node bifurcation ($\dot{x} = r + x^2$). Two fixed points are created spontaneously at the origin as r is increased through $r > r_c$, with a semi-stable fixed point at $x = 0$ for the special case $r = r_c$.

c.



$$5 - r e^{-x^2} = 0$$

$$e^{-x^2} = \frac{5}{r}$$

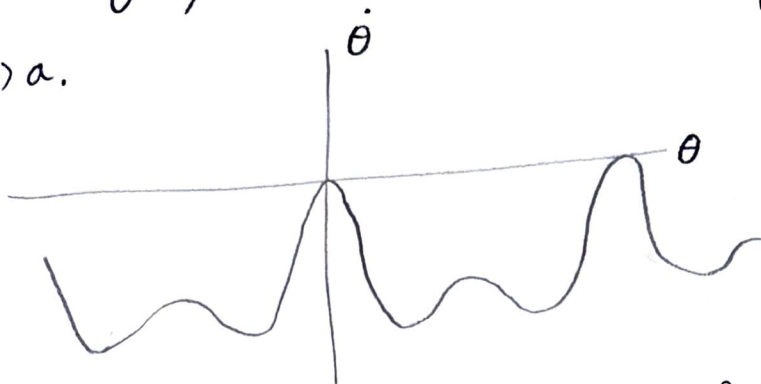
$$-x^2 = \ln \frac{5}{r}$$

$$\Rightarrow x^* = \pm \sqrt{-\ln\left(\frac{5}{r}\right)}$$

since $\frac{5}{r} < 1$ for $r > r_c$, this expression is defined. It is clearly imaginary for $r < r_c$.

2. $\dot{\theta} = \mu + \cos \theta + \cos 2\theta$ if $\mu < -2$:

(2.1) a.



$\mu = -2$:



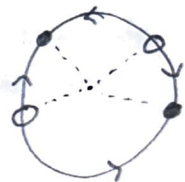
$-2 < \mu < 0$



$\mu = 0$:



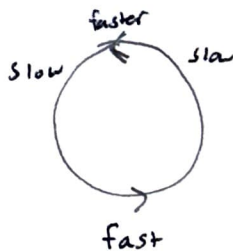
$0 < \mu < \frac{9}{8}$:



$\mu = \frac{9}{8}$



$\mu > \frac{9}{8}$



$\text{Min } \dot{\theta} = \mu + \text{Min}(\cos \theta + \cos 2\theta)$

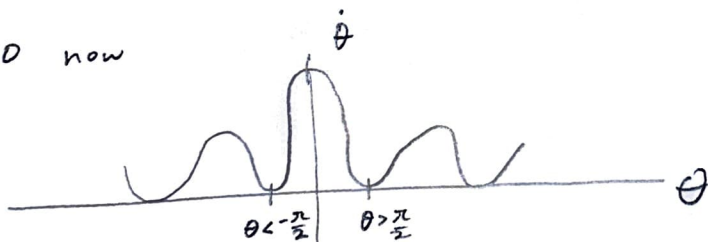


$\mu = -2$



$\mu = 0$

$\dot{\theta} \geq 0$ now



$\mu = \frac{9}{8}$

2. b. There are 3 separate bifurcations that occur in this system, as shown in the above sketches:

(2.2)
(and
(2.3))

1. Saddle node / Blue sky bifurcation at $\mu = -2$

as μ crosses $\mu_c = -2$ threshold, the system goes from no fixed points to a stable & unstable fixed point to the right and left of $\theta = 0$, stemming from a semi-stable fixed point @ $\mu = -2$.
Of course, $\cos \theta + \cos 2\theta = 2$ at $\theta = 0$.

2. Saddle node / Blue sky bifurcation at $\mu = 0$

as μ changes from negative to positive, there is another pair of stable / unstable fixed points created at $\theta = \pi$.

3. Saddle node bifurcation at $\mu = 9/8$

The two pairs of fixed points each collide and annihilate, leaving a system where $\dot{\theta} > 0$ for all θ if $\mu > 9/8$.

The calculation of $\mu = 9/8$ is given below:

C.
2.3 To calculate the value of $\mu = 9/8$, we let CMH (3) (4)
(cont)

$$\min \dot{\theta} = \mu + \min (\cos \theta + \cos 2\theta)$$

$$\hookrightarrow \frac{d}{d\theta} (\cos \theta + \cos 2\theta) = -\sin \theta - 2\sin 2\theta = 0$$

$$\Rightarrow \sin \theta = -2 \sin 2\theta$$

$$\Rightarrow \sin \theta = -4 \sin \theta \cos \theta$$

$$\cos \theta = -\frac{1}{4}$$

$$\Rightarrow \boxed{\theta_{\min} = \cos^{-1}\left(-\frac{1}{4}\right)}$$

Evaluating

$$\dot{\theta}(\theta_{\min}) = \mu + \cos(\arccos(-\frac{1}{4})) + \cos(2\arccos(-\frac{1}{4})) :$$

$$= \mu + -\frac{1}{4} - \frac{1}{2} - \frac{3}{8}$$

(using $\cos 2x = \cos^2 x - \sin^2 x$)

$$\Rightarrow \dot{\theta}(\theta_{\min}) = \mu - 9/8$$

Hence, @ $\mu = \mu_c = 9/8$, $\dot{\theta}$ loses its only remaining fixed points.

3.1 a) Equilibria occur when $\dot{x} = \dot{y} = 0$. Hence,

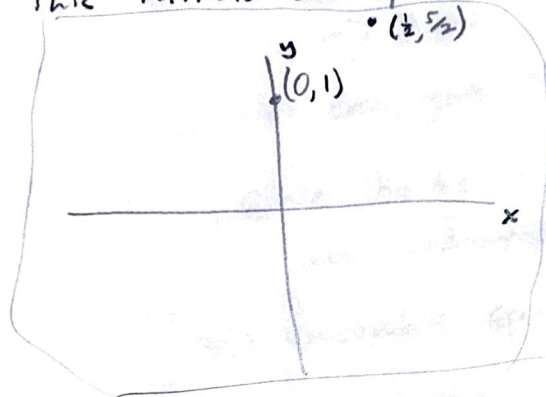
$$\left. \begin{aligned} \dot{x} &= x(3-y-2x^2) \\ \dot{y} &= (y-1)(x-\frac{1}{2}) \end{aligned} \right\} = 0$$

$$\dot{x} = 0 \text{ for } x=0, \text{ or for } y = 3-2x^2 \quad \text{if } x = \frac{1}{2}, \quad y = 3 - \frac{1}{2} = 2.5$$

$$\dot{y} = 0 \text{ for } y=1, \quad x = \frac{1}{2}$$

$$\text{Equilibria } (x^*, y^*) \rightarrow (0, 1); (\frac{1}{2}, \frac{5}{2})$$

Since the only way $\dot{y} = 0$ if $y \neq 1$ is if $x = \frac{1}{2}$, this restricts our equilibria to 2 coordinate pairs.



3.2 b)

$$3.3 \text{ c) } \underline{A} = \begin{pmatrix} \partial_x \dot{x} & \partial_y \dot{x} \\ \partial_x \dot{y} & \partial_y \dot{y} \end{pmatrix} = \begin{pmatrix} 3-y-6x^2 & -x \\ y-1 & x-\frac{1}{2} \end{pmatrix}$$

$$3.4 \text{ d) } \underline{A}(0, 1) = \begin{pmatrix} 3-1-0 & -0 \\ 0 & -1/2 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & -1/2 \end{pmatrix} \equiv \underline{A}_1$$

$$\underline{A}(\frac{1}{2}, \frac{5}{2}) = \begin{pmatrix} 3-\frac{1}{2}-6\cdot\frac{1}{4} & -1/2 \\ 3/2 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1/2 \\ 3/2 & 0 \end{pmatrix} \equiv \underline{A}_2$$

3.5 (e) $A_1 = \begin{pmatrix} 2 & 0 \\ 0 & -1/2 \end{pmatrix} \Rightarrow \text{Tr } A_1 = 3/2, \det A_1 = -1$ CH 6

$$\Rightarrow \lambda_{\pm} = 3/4 \pm 5/4$$

$$\Rightarrow \lambda = 2, -1/2$$

$\Rightarrow$ Saddle point: stable direction has $\lambda = -1/2$,
unstable has $\lambda = 2$

$$A_2 = \begin{pmatrix} 1 & -1/2 \\ 3/2 & 0 \end{pmatrix} \Rightarrow \text{Tr } A_2 = 1, \det A_2 = 3/4$$

$$\Rightarrow \lambda_{\pm} = \frac{1}{2} \pm \frac{i}{\sqrt{2}}$$

complex conjugate roots

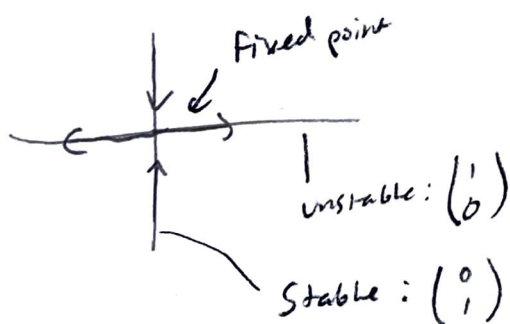
Since $\text{Re } \lambda_{\pm} > 0$, the oscillations
are undamped

$\Rightarrow$ unstable spiral

3 (f)

g.6 $(0, 1)$

eigenvectors: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$



$$\left(\frac{1}{2}, \frac{5}{2} \right)$$

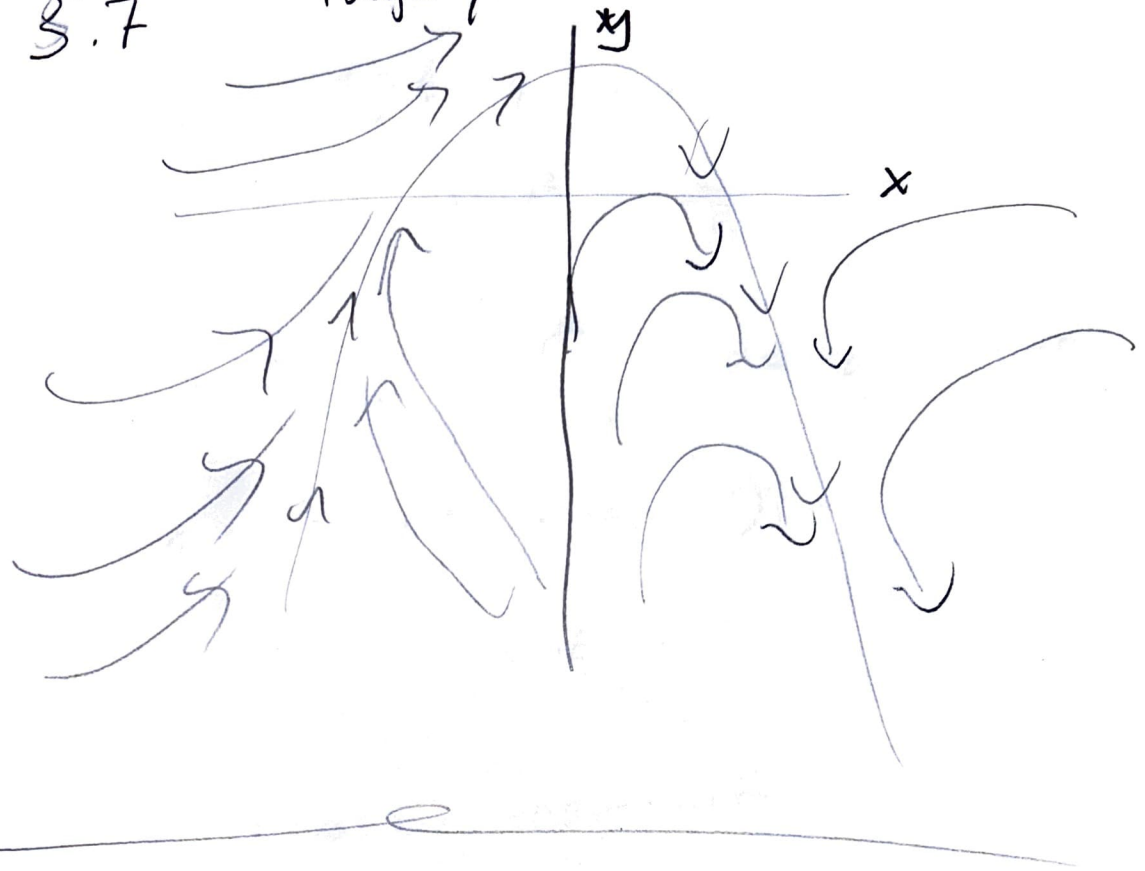
eigenvectors: $\begin{pmatrix} \frac{i}{3}(\sqrt{2} \pm i) \\ 1 \end{pmatrix}$



outward
spiral

§.7

rough qualitative picture



4.1

4.a $\dot{a} = -p + q(1 - q^2 - p^2) \quad \dot{q} = q + p(1 - q^2 - p^2)$

$$r^2 = x^2 + y^2 = q^2 + p^2$$

$$\dot{q} = -p + q(1 - r^2) \quad \dot{p} = q + p(1 - r^2)$$

$$\dot{q} = -r \sin \theta + r \cos \theta (1 - r^2)$$

$$\dot{p} = r \cos \theta + r \sin \theta (1 - r^2)$$

~~$$\dot{q} = \dot{r} \cos \theta + r \sin \theta \dot{\theta} = \dot{r} \cos \theta + r \cos \theta (1 - r^2)$$~~

~~$$\dot{p} = \dot{r} \sin \theta + r \cos \theta \dot{\theta} = \dot{r} \sin \theta + r \sin \theta (1 - r^2)$$~~

~~$$\dot{r} = r(1 - r^2)$$~~

$$\dot{q} = \dot{r} \cos \theta - r \sin \theta \dot{\theta} = -r \sin \theta + r \cos \theta (1 - r^2)$$

$$\dot{p} = \dot{r} \sin \theta + r \cos \theta \dot{\theta} = r \cos \theta + r \sin \theta (1 - r^2)$$